Aiginition Longitudinal Biomarker Investigation Of Neurodegeneration (ALBION)

Research protocol

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Objective

Aiginition Longitudinal Biomarker Investigation Of Neurodegeneration (ALBION) is designed to address research questions concerning the preclinical and prodromal stage of Alzheimer's Disease (AD). A basic methodological principle of the present study, lies in the comprehensive evaluation of study participants, through collection of bio-samples in combination with extensive information obtained from laboratory tests, validated questionnaires, portable devices, and medical examination. All these sources of information, with some of them obtained at multiple time-points, will allow us to specifically address pathways of disease development, providing insights to the biological mechanisms that may explain observed relationships. Determining the preclinical stage of dementia may provide a broader perspective of the changes taking place in AD brains, leading to effective therapeutic and preventive strategies.

Participants

Inclusion and exclusion criteria

<u>Inclusion criteria</u>: People above the age of 40 years old who have a positive family history or report subjective cognitive complaints. Participants are a) cognitively normal people, b) people with subjective cognitive decline or c) people with Mild Cognitive Impairment but clearly, do not meet the criteria for dementia diagnosis.

Thus, <u>exclusion criteria</u> include the following:

1. Diagnosis of dementia
2. Neurological, psychiatric or medical conditions associated with a high risk of cognitive impairment or dementia including but not limited to Huntington's disease, multiple sclerosis, Parkinson's disease, Down syndrome, active alcohol/ drug abuse or major psychiatric conditions including, not limited to, major depressive disorder, schizophrenia and bipolar disorder.
3. MRI contraindications

4. Use of anticoagulant medication

ALBION study procedures & timeline

All participants with normal cognitive function are evaluated every two years after their baseline assessment, but every year we contact with the participants by phone. If a participant is diagnosed with dementia or Mild Cognitive Impairment or asks it during the phone contact, he/she is evaluated annually. Each evaluation includes the measurements described in Table 1 and Figures 1-4.

Participants will be followed up longitudinally for a period of at least 10 years in order to be able to observe potential cognitive changes and disease progression given the fact that neurodegenerative diseases have a long evolutionary course.

Table 1. Procedures included in ALBION study organized per visit

	1st visit	2nd visit[a]	3rd visit	4th visit[a]	5th visit	6th visit[a]	7th visit[c]
Neurological examination (+ CDR, NPI, IADL, HACHINSKI, UPDRS)	X	X	X	X	X	X	X
Neuropsychological assessment	X	X	X	X	X	X	X
HADS questionnaire	X	X[b]	X		X[c]		X
Sleep quality and quantity questionnaire	X	X[b]	X		X[c]		X
Leisure Activities questionnaire	X	X[b]	X		X[c]		X
Subjective Cognitive Complaint Questionnaire (SCC questionnaire)	X	X[b]	X		X[c]		X
Athens Physical Activity questionnaire	X	X[b]	X		X[c]		X
Dynamometer (3 trials)	X	X[b]	X		X[c]		X
24h dietary recall	X	X[b]	X		X[c]		X
Anthropometrics (weight/height)	X	X[b]	X		X[c]		X
Gait speed (2 trials)	X	X[b]	X		X[c]		X
Magnetic resonance imaging (MRI)	X						X
Electroencephalogram (EEG)	X		X		X[c, d]		X
Lumbar Puncture	X						X[e]
Blood sample	X		X		X		X
Stool sample (microbiome analysis)	X		X		X		X
Wrist actigraphy	X		X		X		X
WatchPat	X		X[c, d]		X[c, d]		X

[a] Update 4/2023: These visits will be performed only if a participant is diagnosed with Mild Cognitive Impairment, dementia or if a participant asks it in the contact performed annually by phone

[b]These procedures were performed till August 2021. After that, visit 2 will include only neurological and neuropsychological assessment.

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[C] Update 2023
[d] These assessments will be performed only to those participants that did not perform the respective assessment to the previous visit
[e] Excluded from second lumbar puncture are those 1) who have been diagnosed with dementia in a previous visit and b) those who had a lumbar puncture after participating in other department's study (EVOKE…)

Figure 1: 1st assessment

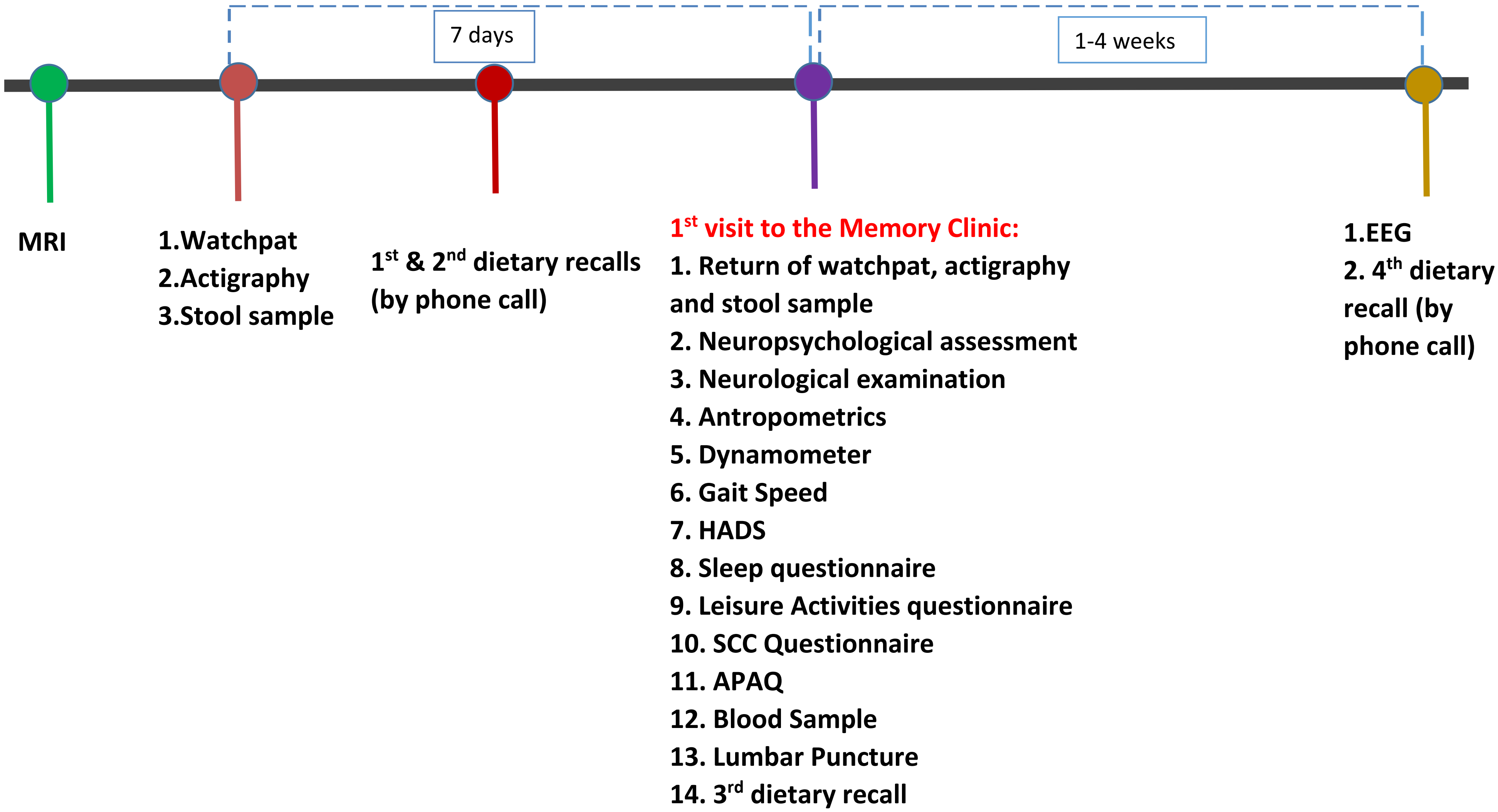


Figure 2: 2nd assessment

* These procedures were performed till August 2021. After that, visit 2 will include only neurological and neuropsychological assessment.

Update 4/2023: This visit will be performed only if a participant is diagnosed with Mild Cognitive Impairment, dementia or if a participant asks it in the contact performed annually by phone.

Figure 3: 3rd assessment

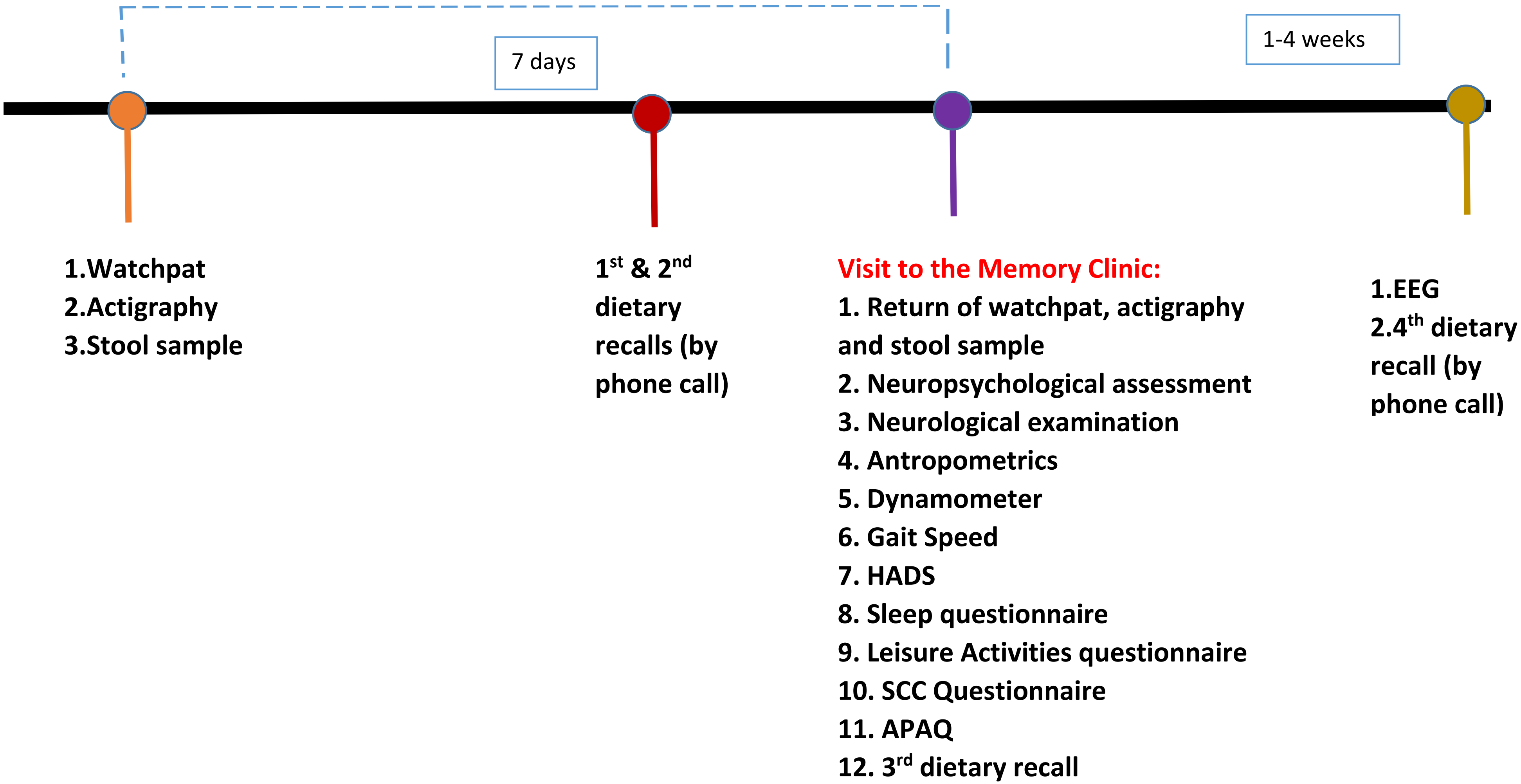
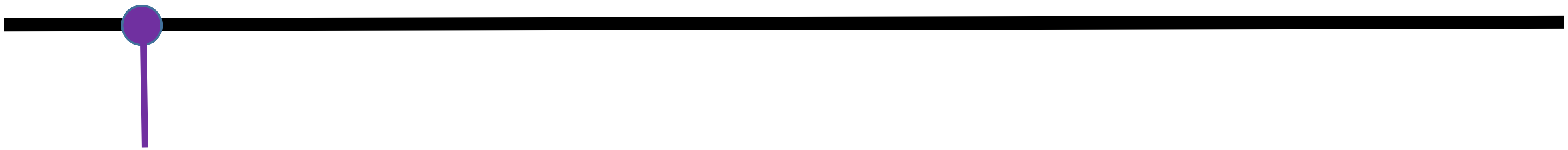


Figure 4: 4th assessment



Visit to the Memory Clinic:

1. Neuropsychological assessment

2. Neurological examination

Update 4/2023: This visit will be performed only if a participant is diagnosed with Mild Cognitive Impairment, dementia or if a participant asks it in the contact performed annually by phone.

Figure 5: 5th assessment

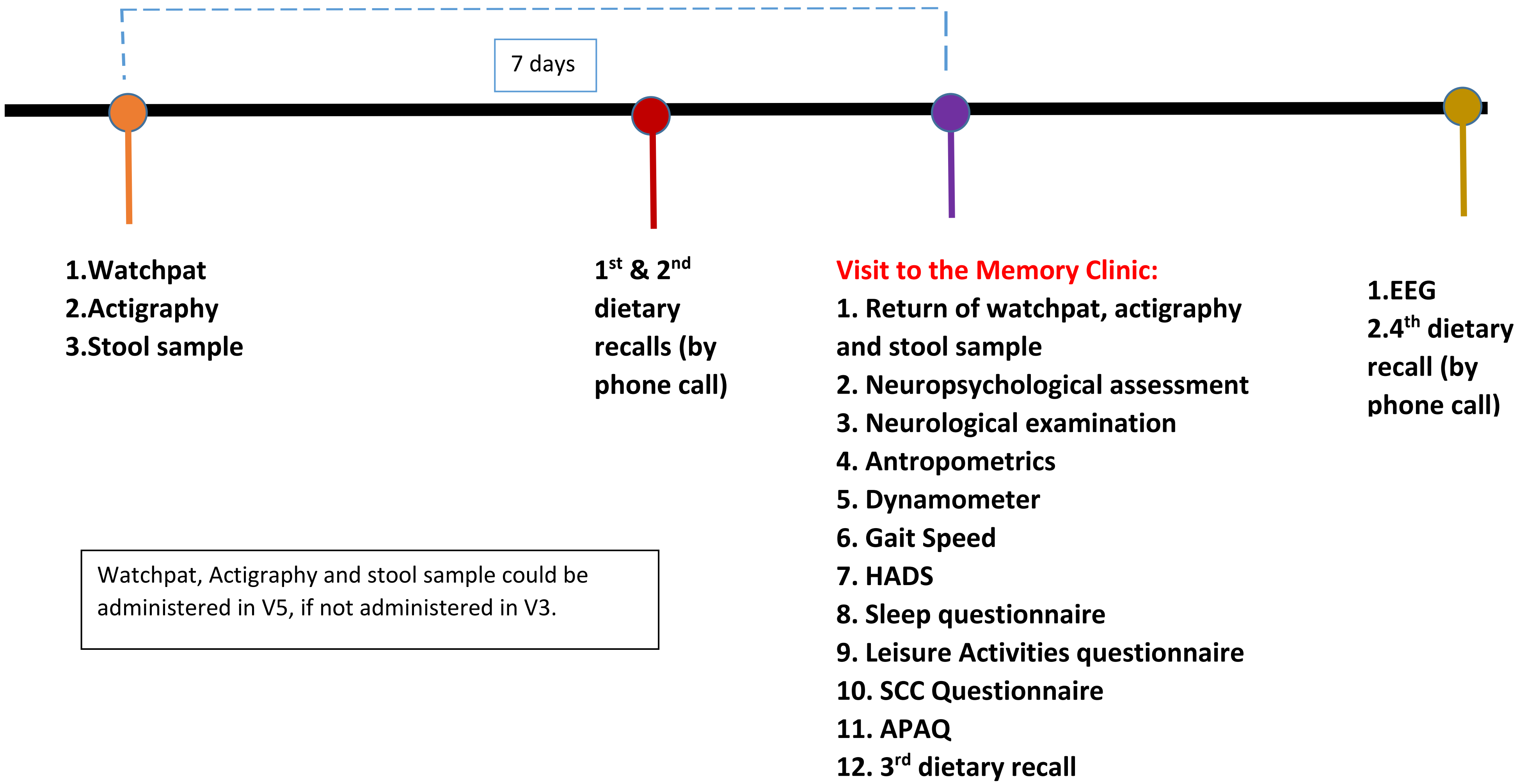


Figure 6: 6^{th} assessment

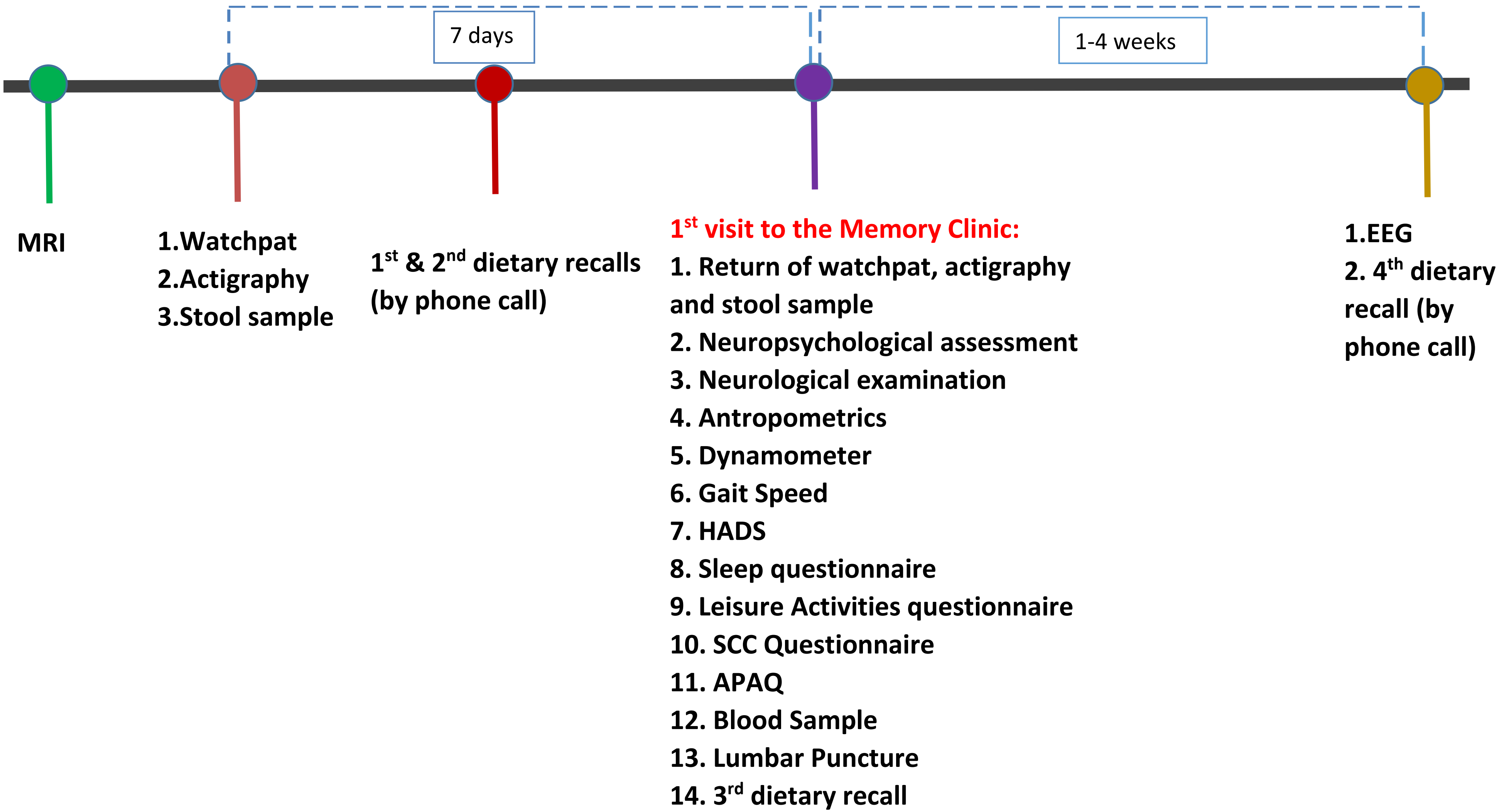
Visit to the Memory Clinic:

1. Neuropsychological assessment

2. Neurological examination

Update 4/2023: This visit will be performed only if a participant is diagnosed with Mild Cognitive Impairment, dementia or if a participant asks it in the contact performed annually by phone.

Figure 7: 7th assessment



MRI

All participants undergo a whole-brain imaging on a 3T Philips Achieva-Tx MR scanner (Philips, Best, The Netherlands) equipped with an eight-channel head coil. The imaging protocol contains basic clinical and research used sequences. The basic protocol consists of a sagittal T2-fluid attenuation inversion recovery (T2-FLAIR), an axial T2-TSE combined with fat suppression and an axial T2* gradient echo to exclude severe cerebrovascular disease. Prior to data analyses, MRI data will be reviewed by a neuroradiologist to exclude any cerebrovascular disease and by an MRI physicist to identify system or patient-related artifacts. Research protocol includes:

- A sagittal high-resolution 3D T1 weighted to estimate the gray matter volumes and the cortical thickness of different brain areas (acquisition voxel size 1.11 × 1.11 x 1.2 mm; TR = 6.7 ms; TE = 3.1 ms; TI = 2500 ms; flip angle = 9°)
- An axial single-shot spin-echo echo-planar imaging (EPI) DTI acquisition with 48 diffusion encoding directions to evaluate the white matter integrity (acquisition voxel size 2 x 2 × 2 mm; 60 slices; two b factors with 0 s/ mm2 (low b) and 1000 s/mm2 (high b); sensitivity encoding reduction factor of 2; TR = 7576 ms; TE = 91 ms)
- A B0 reversed polarity gradient compared to the previous DTI acquisition using the same acquisition parameters to correct the geometrically distorted images due to EPI, during the post-processing process.
- An axial 3D susceptibility-weighted imaging (SWI) (acquisition voxel size 0.60 × 0.60 x 2.00 mm; TR = 29 ms; TE = 25 ms; flip angle 15°; number of echoes) to identify and quantify possible blood deficits or calcium. POSTGRADUATE MEDICINE 503
- An axial T2* weighted gradient echo combined with EPI for resting-state functional MRI (rs-fMRI) (acquisition voxel size 3 x 3 x 3 mm2 , 38 slices; TR = 1900 ms; TE = 30 ms; flip angle = 80°, dynamics = 204) to evaluate the functional connectivity between different brain regions. During the rs-fMRI scan participants are instructed to lie still with their eyes closed.
- An axial T2 weighted spin echo combined with EPI (SE-EPI) rs-fMRI (acquisition voxel size 3 × 3 × 3 mm2 , 38 slices; TR = 4000 ms; TE = 60 ms; flip angle = 80°) to evaluate the functional connectivity between brain regions which are imaged

distorted due to different chemical susceptibilities of the neighbored tissues (e.g. temporal lobe).

- The previous SE-EPI fMRI sequence will be repeated using reversed gradient polarity and reduced dynamic scans to correct the geometrically distorted images due to EPI, during the post-processing process.
- An axial pseudo-continuous Arterial Spin Labeling (pcASL) sequence to estimate quantitatively the tissue perfusion without contrast agent usage (acquisition voxel size 3.44 × 3.58 x 4.50 mm; 36 slices; TR = 4943 ms; TE = 10 ms; 30 dynamics; EPI factor = 25; flip angle = 90°).
- M0 scans using TR = 2s and 10s using the same acquisition parameters of pcASL sequence except of TR are acquired to improve the signal of M0.

Based on the results and the needs of our research additional MRI sequences may be added to the existing protocol in the future.

Wrist Actigraphy

Actiwatch 2 (Phillips-Respironics) is used to record sleep parameters (Picture 1). For product specifications visit: https://www.usa.philips.com/healthcare/product/HC1044809/actiwatch-2-activity-monitor/specifications

Picture 1. The Actiwatch 2 (Phillips-Respironics) used in ALBION study

The wrist actigraph is given to participants 7 days before their visit to the Memory Clinic. They are advised to wear the actigraph in the non-dominant hand for seven days and nights (except bath time) and in the meanwhile they keep a daily sleep diary, shown below:

Name: ……………………………………………………… ID: ………….. Assessment: ……………………………………………

Date	Time I lied down	Time I switched off the lights	Time I woke up	Time I got up	Medications		Please record if you had any naps (during the daytime)	
					medications	time each medications is consumed	time I slept	time I woke up

Picture 2. Sleep diary kept during actigraphy

After the export of the data, an actigram is acquired for each participant (as shown in Picture 3) and a large number of variables regarding sleep parameters are saved in a data file.

Watchpat

The WatchPAT can provide a reasonably accurate measurement of various sleep parameters, and may be a clinically reliable tool for diagnosing Obstructive Sleep Apnea (OSA) (https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6175781/). In this study the WatchPAT™300 device has been used (Picture 4). Details can be found in: https://www.itamar-medical.com/professionals/watchpat-300/.

Picture 3. The WatchPat 300 device.

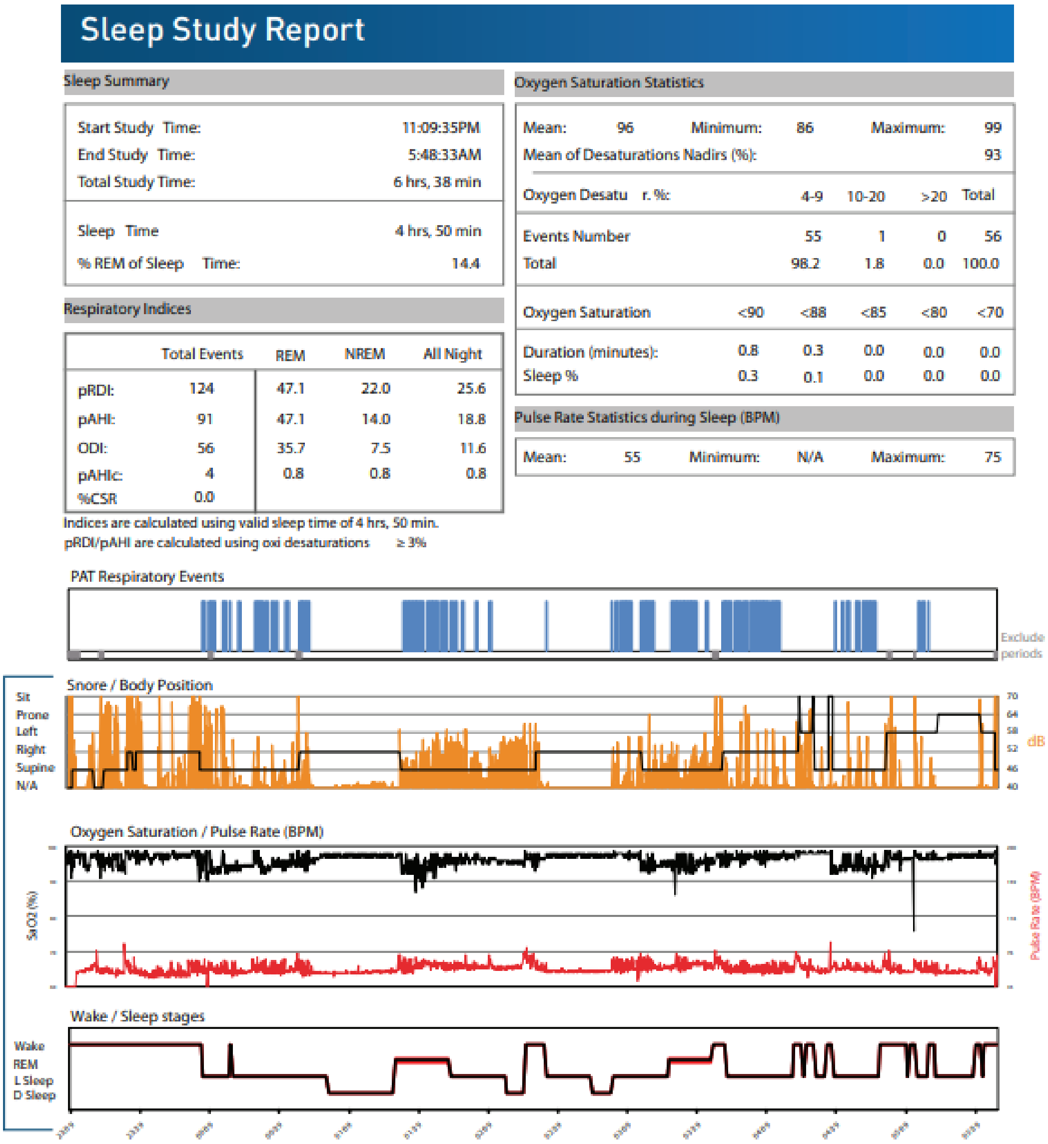
Picture 4. The WatchPat 300 device.

Participants take the WatchPAT™300 the same day with the actigraph. However, they are advised to use it the night before their visit to the Memory Clinic (and thus the last night of the actigraphy records). They are also given detailed written and illustrated instructions regarding the proper use of the WatchPat.

The WatchPAT™300 procedure is omitted, if one of the following is present (exclusion criteria):

1. Use of one of the following medications: alpha blockers, short acting nitrates (less than 3 hours before study initiation).
2. Permanent pacemaker: atrial pacing or VVI without sinus rhythm.
3. Sustained* non-sinus cardiac arrhythmias. (In the setting of sustained arrhythmia the WatchPAT's automated algorithm might exclude some periods of time, resulting in a reduced valid sleep time. A minimum valid sleep time of 90 minutes is required for an automated report generation).

Watchpat data export include an automated report (Picture 5) and more than 80 variables regarding sleep parameters.



Sleep Study Report

Sleep Summary

Start Study Time:	11:09:35PM
End Study Time:	5:48:33AM
Total Study Time:	6 hrs, 38 min
Sleep Time	4 hrs, 50 min
% REM of Sleep Time:	14.4

Respiratory Indices

	Total Events	REM	NREM	All Night
pRDI:	124	47.1	22.0	25.6
pAHI:	91	47.1	14.0	18.8
ODI:	56	35.7	7.5	11.6
pAHIc:	4	0.8	0.8	0.8
%CSR	0.0			

Indices are calculated using valid sleep time of 4 hrs, 50 min.
pRDI/pAHI are calculated using oxi desaturations ≥ 3%

Oxygen Saturation Statistics

Mean:	96	Minimum:	86	Maximum:	99
Mean of Desaturations Nadirs (%):					93

Oxygen Desatu r. %:	4-9	10-20	>20	Total
Events Number	55	1	0	56
Total	98.2	1.8	0.0	100.0

Oxygen Saturation	<90	<88	<85	<80	<70
Duration (minutes):	0.8	0.3	0.0	0.0	0.0
Sleep %	0.3	0.1	0.0	0.0	0.0

Pulse Rate Statistics during Sleep (BPM)

Mean:	55	Minimum:	N/A	Maximum:	75

Picture 5. Automated report produced by WatchPat.

Participants receive a detailed report with the WatchPat results after successful completion of the sleep study, and those with an Apnea Hypopnea Index (AHI) greater than 15, are referred to a Sleep Clinic for further investigation and treatment.

Stool sample

Participants are given the necessary stool sample test kit in order to ensure sterilized conditions of the procedure as well as the right storage conditions until sample delivery to our center. This kit contains the following and is depicted in Picture 6:

- 2 collection and preservation tubes (they will be used in the same fecal sample), marked with the unique ID number of each participant.
- collection paper
- ice pack
- a pair of single use plastic gloves
- small bag for freezer storage
- Bristol stool scale (to be completed by the participants)
- Written and illustrated instructions regarding the use of the stool test kit

Picture 6: Stool sample kit

Concurrent use of oral antidiabetic medications and the use of antibiotics up to two months before the fecal sample collection is recorded for all participants. The fecal sample collection is performed maximum 2 days (48h) before participants visit the Memory Clinic. When received, the stool samples are stored in freezers in -80oC. Stool smples are coded as follows: 4-digit participant's code - numbers that identify the number of sample collection. Regarding numbers, 1 stands for first time of stool collection, 2 stands for second time of collection, etc.

24h Dietary recalls

Dietary intake is assessed through four 24-h dietary recalls; some of them administered on-site and some by telephone. Three of the recalls refer to weekdays and one to a weekend day. Specifically, trained dieticians ask the participants to report in detail all foods and beverages consumed during the previous day, defined as the timeframe between waking up in the morning and going to bed at night, using the multiple pass approach. Recall data are analyzed in terms of nutrients using the dietary analysis software Nutritionist Pro™ (2007, Axxya Systems, Texas, USA). Additionally, dietary intake is grouped into food groups, including dairy products, cereals, fruits, vegetables, meat, fish, legumes, added fats, alcoholic beverages, stimulants, and sweets. Furthermore, participants are asked about the water intake during the previous day, whether they use a dietary supplement and whether they are in a weight loss diet. Finally, for each eating episode, information about time, location, concurrent activities, and eating companion is recorded.

Lumbar Puncture

The procedure of LP as well as the collection, processing and storage of CSF is conducted according to international guidelines. CSF is primarily processed for biomarkers such as Aβ42, T-tau, and P-tau, indicative of Alzheimer disease.

LP procedure

- Time of LP: weekdays, morning time (from 09:00 to 13:00)
- Fasting: no
- Volume of CSF collected: 10ml
- Location: Intervertebral space L3–L5
- Type of needle: Atraumatic 25G or other LP needle 20g
- Type of collection tube: pp tubes
- Blood contamination: in case CSF is blood contaminated, the procedure is repeated
- CSF analysis: protein and glucose levels, cell counts with differential, color, microscopic examination
- Centrifugation: 2000 g for 10 min at RT
- Aliquots: After centrifugation, 15 aliquots will be filled with 500μL of CSF
- Storage temperature: -80°C
- Coding of samples: 4-digit participant's code- CSF – numbers that identify the number of sample collection. Regarding numbers, 1 stands for first time sample

collection, 2 stands for second time sample collection, etc. Example: 6742-CSF-1 refers to participant's 6742 first CSF sample collection.

Till now, measurements of CSF neurodegenerative biomarkers has been performed in three different labs: 1) Aristotle University (using ELISA), b) Aiginiteio Hopsital (using ELISA), c) University of Gothenburg (from EPAD, using Roche), d) University of Larisa (using Roche), while several aliquots of CSF are stored for future use in cryoboxes in Aiginiteio Hospital and Harokopeio University.

Aristotle University

The analysis is performed with Stat Fax 2100 ELISA microplate reader. The kits include the following:

- INNOTEST h TAU protein, CE/IVD (FUJIREBIO / INNOGENETICS Belgium, 96 tests/kit [81572] – more information: https://www.fujirebio.com/en/products-solutions/innotestr-htau-ag
- INNOTEST Phospho TAU protein, CE/IVD (FUJIREBIO / INNOGENETICS Belgium, 96 tests/kit [81574]) - more information: https://www.fujirebio.com/en/products-solutions/innotestr-phosphotau181p
- INNOTEST β-AMYLOID (1-42) kit (FUJIREBIO / INNOGENETICS Belgium, 96 tests/kit [81576])- more information: https://www.fujirebio.com/en/products-solutions/innotestr-bamyloid142-0

Abnormal values using this method:

- Aβ<480 pg/ml
- Tau>195 pg/ml
- PTau>61 pg/ml

University of Gothenburg

The analysis is performed using the Roche Diagnostics Elecsys Platform. Using this system a threshold of 1,000pg/ml of ABeta42 was agreed upon to define amyloid positivity. Regarding PTau and total Tau, a value above 27pg/mL and a value above 300pg/mL respectively are considered abnormal. Moreover, abnormal values are considered the following:

• If pTau/Abeta42 ratio >0.024 test result **positive** *(If pTau/Abeta42 ratio ≤0.024 test result negative).*

• If tTau/Abeta42 ratio >0.28 test result **positive** *(If tTau/Abeta42 ratio ≤0.28 test result negative).*

- For more information can be found in the links below: https://diagnostics.roche.com/global/en/products/params/elecsys-abeta42.html
- https://diagnostics.roche.com/global/en/products/params/elecsys-ttau.html
- https://diagnostics.roche.com/global/en/products/params/elecsys-ptau.html

Aiginiteio Hospital

The analysis is performed with Stat Fax 2100 ELISA microplate reader

- Total-Tau ELISA kit (Product No. 6531-L, EUROIMMUN Medizinische Labordiagnostika AG, Lübeck, Germany)- more information: https://www.euroimmun.de/en/products/productdetails/6531-L/2/74074/
- INNOTEST β-AMYLOID(1-42) ELISA kit (Product No. 81576, Fujirebio Europe N.V., Gent, Belgium) – more information: https://www.fujirebio.com/en/products-solutions/innotestr-bamyloid142-0
- INNOTEST PHOSPHO-TAU(181P) ELISA kit (Product No. 81574, Fujirebio Europe N.V., Gent, Belgium)-more information: more information: https://www.fujirebio.com/en/products-solutions/innotestr-phosphotau181p

Abnormal values using this method:

- Aβ<565 pg/ml
- Tau >452 pg/ml
- Ptau>66,26 pg/ml

University of Larisa

The analysis is performed using the Roche Diagnostics Elecsys Platform. Using this system a threshold of 1,000pg/ml of ABeta42 was agreed upon to define amyloid positivity.

Regarding PTau and total Tau, a value above 27pg/mL and a value above 300pg/mL respectively are considered abnormal. Moreover, abnormal values are considered the following:

• If pTau/Abeta42 ratio >0.024 test result **positive** *(If pTau/Abeta42 ratio ≤0.024 test result negative).*

• If tTau/Abeta42 ratio >0.28 test result **positive** *(If tTau/Abeta42 ratio ≤0.28 test result negative).*

- For more information can be found in the links below: https://diagnostics.roche.com/global/en/products/params/elecsys-abeta42.html
- https://diagnostics.roche.com/global/en/products/params/elecsys-ttau.html

https://diagnostics.roche.com/global/en/products/params/elecsys-ptau.html

Blood Sample

Blood sampling procedure

- Date: blood sample is collected the same day with the LP, if a LP is included in the visit
- Fasting: no
- Volume of blood: 35 ml
- Necessary materials: 5 potassium EDTA tubes, 2 Serum tubes with gel, 1 RNA (stabilization and lyse blood cells) tube, 1 white blood cells isolation tube, 29 Eppendorf tubes (1,5ml)
- Blood fractionation is performed and the subsequent component parts are analyzed as follows:

  Plasma and buffy coat

  Blood sample is collected in 5 potassium EDTA tubes and centrifugated in 2000g for 10 minutes. After centrifugation, blood plasma is aliquoted in 10 Eppendorf tubes (200μl/Eppendorf) and buffy coat in 4 (200μl/Eppendorf).

  Serum

  Blood is collected in 2 serum tubes with gel and centrifugated in 2000g for 10 minutes. After centrifugation, the serum is aliquoted in 15 Eppendorf tubes (400μl/Eppendorf)

Leukocytes

3 mL of heparinized blood are obtained from each volunteer. This sample is left in 4°C till transfer to Harokopeio University where blood will be further processed. Specifically, in order to induce erythrocyte sedimentation, 1 mL of dextran solution (3% dextran in NaCl 0.15 M) is added and the mixture is kept for 1h at room temperature. The leukocyte rich supernatant is then centrifuged at 500 x g for10 min at room temperature. Contaminating erythrocytes of the sediment are lysed with the addition of 3 ml lysis solution consisting of 155 mM NH_4Cl, 10 mM $KHCO_3$ and 0.1 mM EDTA and then removed with a centrifugation at 300 x g for 10 min at room temperature. The pelleted cells are resuspended in 0.6 mL of a buffer containing 50 mM Tris-HCl (pH 7.4), 0.25 M sucrose and 1 mM DTT and then sonicated on ice for 4 times of 10 sec each.

The leukocyte homogenate is aliquoted in 3 Eppendorf tubes (200μl)/ Eppendorf) and stored at −80 ◦C.

RNA

3ml blood is collected in Tempus™ Blood RNA tubes, containing 6ml RNA stabilization solution (guanidine thiocyanate). More information can be found in: https://www.fishersci.co.uk/shop/products/tempus-blood-rna-tube/12063745#:~:text=Tempus%E2%84%A2%20Blood%20RNA%20Tubes,reagent%2C%20lysis%20occurs%20almost%20immediately.

- Samples are coded as follows:
    - For RNA sample: 4-digit participant's code – numbers that identify the number of sample collection. Regarding numbers, 1 stands for first time sample collection, 2 stands for second time sample collection, etc. Example: 6742-1 refers to participant's 6742 first RNA sample collection.
    - For other blood samples: 4-digit participant's code- letters that identify type of sample – numbers that identify the number of sample collection. Specifically, S stands for serum, P stands for plasma and buffy is for buffy coat. Regarding numbers, 1 stands for first time sample collection, 2 stands for second time sample collection, etc. Example: 6742-S-2 refers to participant's 6742 second Serum sample collection.

- Storage temperature: -80°C
- Genetic outcomes: APOE genotyping is performed in genomic DNA extracted from blood buffy coat, using Qiamp DNA Blood Mini Kits (Qiagen, Venlo, Netherlands). The method used for the genotyping was polymerase chain reaction-DNA sequencing, carried out with LightCycler 2 (Roche Diagnostics GmbH) and using the LightMix TIB MOLBIOL reactors.

Storage of biological samples

The cryoboxes containing the Eppendorf tubes with CSF and blood samples are stored, half in Aiginiteio Hospital and half in Harokopeio University. In order to keep track of the cryoboxes content, detailed logs are recorded with the exact position of each Eppendorf (Table 2). These logs are saved both in hard copies and in e-files.

Table 2. Cryobox log example

Number of box	1	2	3	4	5	6	7	8	9	10
A	6042 S-1	6042 S-1	6042 S-1	6042 S-1	6042 P-1	6042 P-1	6042 P-1	6042 BUFFY-1	6042 CSF-1	6042 CSF-1
B	6050 CSF-1	6050 CSF-1	6050 S-1	6050 S-1	6050 S-1	6050 P-1	6050 P-1	6050 P-1	6074 S-1	6074 S-1
C	6074 P-1	6074 P-1	6074 P-1	6074 P-1	6070 CSF-1	6070 CSF-1	6070 CSF-1	6051 S-1	6051 S-1	6051 S-1
D	6051 S-1	6051 CSF-1	6051 CSF-1	6051 CSF-1	6051 P-1	6051 P-1	6051 P-1	6060 BUFFY-1	6060 P-1	6060 P-1
E	6060 CSF-1	6060 CSF-1	6060 CSF-1	6060 S-1	6060 S-1	6060 S-1	6072 BUFFY-1	6072 P-1	6072 P-1	6072 S-1
F	6072 CSF-1	6072 CSF-1	6072 CSF-1	6072 S-1	6072 S-1	6068 CSF-1	6068 CSF-1	6068 BUFFY-1	6068 P-1	6068 P-1
G	6068 S-1	6068 S-1	5902 BUFFY-1	5902 P-1	5902 P-1	5902 P-1	5902 CSF-1	5902 CSF-1	5902 CSF-1	5902 S-1
H	5902 S-1	5902 S-1	6087 S-1	6087 S-1	6087 S-1	6087 S-1	6087 P-1	6087 P-1	6087 CSF-1	6087 CSF-1
I	6087 CSF-1	6058 P-1	6058 P-1	6058 P-1	6058 CSF-1	6058 CSF-1	6058 CSF-1	6058 S-1	6058 S-1	6058 S-1
J	6093 P-1	6093 P-1	6093 CSF-1	6093 CSF-1	6093 CSF-1	6093 S-1	6093 S-1	6093 S-1	6090 P-1	6090 P-1

Four-digit number: patient's ID, S=serum, P=plasma. The last number indicates the number of blood collection number.

Neuropsychological Evaluation

Trained psychometricians administer a battery of neuropsychological tests assessing five cognitive domains: memory, language, attention-speed, executive functioning, and visuospatial perception. The neuropsychological tests used in our study are presented in Table 3.

Table 3. Neuropsychological evaluation	
Cognitive domains	Cognitive test
Global	• Mini-Mental State Examination • Addenbrooke's Cognitive Examination Revised (ACE-R)
Intelligence, pre-morbid level	• Vocabulary
Attention and information processing speed	• Trail-making test part A • Digit Span Forwards
Executive functioning	• Trail-making test part B • Stroop Test • Digit Span Backwards
Visuoperceptual ability	• Medical College of Georgia Complex Figure Test - copy condition • Visuospatial component of ACE-R
Memory	
Verbal memory	• Greek verbal learning test (immediate and delayed recall)] • Story recall (immediate and delayed)
Nonverbal memory	• Medical College of Georgia Complex Figure Test (immediate and delayed recall)
Language	• Semantic and phonological verbal fluency component of ACE-R • Language component of ACE-R • 40-item naming test

Participants' raw scores are transformed to z- scores. Specifically, z-scores were derived for **each** test, using the means and standard deviations (SD) calculated from the scores of the nondemented, non-MCI participants of the ALBION study. Subsequently, z-scores of individual neuropsychological tests were averaged to produce domain z-scores in memory, language, attention-speed, executive and visual-spatial functioning. Decision on grouping of neuropsychological tests was based on a priori neuropsychological knowledge of particular cognitive functions that each test reflects (see Table 3). Furthermore, domain z-scores were averaged in order to calculate a composite z-score. A higher z-score indicates better cognitive performance.

Neurological Examination

Information recorded include demographics, medical history, medications, family history, neurological examination and completion of the following scales: Neuropsychiatric Inventory, HACHINSKI, UPDRS, IADL and CDR (presented below).

Hachinski Ischemic Score

[Hachinski VC, Iliff LD, Zilkha E, et al: Cerebral blood flow in dementia. Arch Neurol 32:632-637, 1975]
[Rosen WG, Terry RD, Fuld PA, Katzman R, Peck A. Pathological verification of ischemic score in differentiation of dementias. Ann Neurol. 1980;7(5):486-8.]

1.Abrupt onset	2
2.Stepwise deterioration	1
3.Somatic complaints	1
4.Emotional Incontinence	1
5.History of hypertension	1
6.History of strokes	2
7.Focal neurological symptoms	2
8.Focal neurological signs	2
*Items 7 and 8refer to symptoms and signs with cerebrovascular origins; for example aphasia due to stroke would be included here. Aphasia such as PPA would not be reflected in the Hachinski Ischemic Score.	

Instrumental Activities Of Daily Living

[Lawton MP, Brody EM. Assessment of older people: self-maintaining and instrumental activities of daily living. Gerontologist. 1969;9(3):179-86.]

IADL TELEPHONE	1= Operates telephone on own initiative; looks up and dials numbers 2= Dials a few well-known numbers 3= Answers telephone, but does not dial 4=Does not use telephone at all
IADL SHOPPING	1=Takes care of all shopping needs independently 2= Shops independently for small purchases 3= Needs to be accompanied on any shopping trip 4=Completely unable to shop
IADL MEALS	1=Plans, prepares, and serves adequate meals independently 2=Prepares adequate meals if supplied with ingredients 3=Heats and serves prepared meals or prepares meals but does not maintain adequate diet 4=Needs to have meals prepared and served
IADL HOUSEKEEPING	1=Maintains house alone with occasion assistance (heavy work) 2=Performs light daily tasks such as dishwashing, bed making 3=Performs light daily tasks, but cannot maintain acceptable level of cleanliness 4=Needs help with all home maintenance tasks 5=Does not participate in any housekeeping tasks
IADL LAUNDRY	1=Does personal laundry completely 2=Launders small items, rinses socks, stockings, etc 3=All laundry must be done by others
IADL TRANSPORTATION	1=Travels independently on public transportation or drives own car 2=Arranges own travel via taxi, but does not otherwise use public transp. 3=Travels on public transportation when assisted/accompanied by another 4=Travel limited to taxi or automobile with assistance of another 5=Does not travel at all
IADL MEDICATIONS	1= Is responsible for taking medication in correct dosages at correct time 2=Takes responsibility if medication is prepared in advance in separate dosages 3=Is not capable of dispensing own medication
IADL MONEY	1=Manages financial matters independently (budgets, writes checks, pays rent and bills, goes to bank); collects and keeps track of income 2=Manages day-to-day purchases, but needs help with banking, major purchases, etc 3=Incapable of handling money
IADL TOTAL	

Neuropsychiatric Inventory (NPI)

SYMPTOM	**NO**	**YES**	**FREQUENCY** **1 2 3 4**	**SEVERITY** **1 2 3**	**ITEM SCORE**
DELUSIONS					
HALLUCINATION					
AGITATION/AGGRESSION					
DEPRESSION					
ANXIETY					
EUPHORIA					
APATHY					
DISINHIBITION					
IRRITABILITY					
ABERRANT MOTOR BEHAVIOR					
NIGHTTIME BEHAVIOR					
APPETITE & EATING					

Clinical Rating Scale (CDR)

[Morris J.S. (1993) The Clinical Dementia Rating: Current version and scoring rules. Neurology 43:2412-2414]

	0	0.5	1	2	3
Memory	No memory loss or slight inconsistent forgetfulness	Consistent slight forgetfulness; partial recollection of events; "benign" forgetfulness	Moderate memory loss; more marked for recent events; defect interferes with everyday activities	Severe memory loss; only highly learned material retained; new material rapidly lost	Severe memory loss; only fragments remain
Orientation	Fully oriented	Fully oriented except for slight difficulty with time relationships	Moderate difficulty with time relationships; oriented for place at examination; may have geographic disorientation elsewhere	Severe difficulty with time relationships; usually disoriented to time, often to place	Oriented to person only
Judgement and problem solving	Solves everyday problems & handles business & financial affairs well; judgment good in relation to past performance	Slight impairment in solving problems, similarities, and differences	Moderate difficulty in handling problems, similarities, and differences; social judgment usually maintained	Severely impaired in handling problems, similarities, and differences; social judgment usually impaired	Unable to make judgments or solve problems
Community affairs	Independent function at usual level in job, shopping, volunteer and social groups	Slight impairment in these activities	Unable to function independently at these activities although may still be engaged in some; appears normal to casual inspection	No pretense of independent function outside home .Appears well enough to be taken to functions outside a family home	No pretense of independent function outside home. Appears too ill to be taken to functions outside a family home
Home and hobbies	Life at home, hobbies, and intellectual interests well maintained	Life at home, hobbies, and intellectual interests slightly impaired	Mild but definite impairment of function at home; more difficult chores abandoned; more complicated hobbies and interests abandoned	Only simple chores preserved; very restricted interests, poorly maintained	No significant function in home
Personal care	Fully capable of self-care		Needs prompting	Requires assistance in dressing, hygiene, keeping of personal effects	Requires much help with personal care; frequent incontinence

Unified Parkinson Disease Rating Scale (UPDRS)

[Fahn S, Elton RL, UPDRS Development Committee. The Unified Parkinson's Disease Rating Scale. In Fahn S, Marsden CD, Calne DB, Goldstein M, eds. Recent developments in Parkinson's disease. Vol. 2. Florham Park, NJ: Macmillan Healthcare Information, 1987:153-163, 293-304]

Speech
0 = Normal.
1 = Slight loss of expression, diction and/or volume.
2 = Monotone, slurred but understandable; moderately impaired.
3 = Marked impairment, difficult to understand.
4 = Unintelligible.
Facial Expression
0 = Normal.
1 = Minimal hypomimia, could be normal "Poker Face".
2 = Slight but definitely abnormal diminution of facial expression.
3 = Moderate hypomimia; lips parted some of the time.
4 = Masked or fixed facies with severe or complete loss of facial expression; lips parted 1/4 inch or more.
Tremor at rest (head, upper and lower extremities)
Head
Left upper extremity
Right upper extremity
Left lower extremity
Right lower extremity
0 = Absent.
1 = Slight and infrequently present.
2 = Mild in amplitude and persistent. Or moderate in amplitude, but only intermittently present.
3 = Moderate in amplitude and present most of the time.
4 = Marked in amplitude and present most of the time.
Action or Postural Tremor of hands
Left upper extremity
Right upper extremity
0 = Absent.
1 = Slight; present with action.
2 = Moderate in amplitude, present with action.
3 = Moderate in amplitude with posture holding as well as action.
4 = Marked in amplitude
Rigidity
Neck
Left upper extremity
Right upper extremity
Left lower extremity
Right lower extremity
0 = Absent.
1 = Slight or detectable only when activated by mirror or other movements.
2 = Mild to moderate.

3 = Marked, but full range of motion easily achieved.
4 = Severe, range of motion achieved with difficulty.

Finger Taps

Left upper extremity
Right upper extremity

0 = Normal. (≥15/5 sec)
1 = Mild slowing and/or reduction in amplitude. (11-15/5 sec)
2 = Moderately impaired. Definite and early fatiguing. May have occasional arrests in movement. (7-10/5 sec)
3 = Severely impaired. Frequent hesitation in initiating movements or arrests in ongoing movement. (3-6/5 sec)
4 = Can barely perform the task.

Hand Movements

Left upper extremity
Right upper extremity

0 = Normal.
1 = Mild slowing and/or reduction in amplitude.
2 = Moderately impaired. Definite and early fatiguing. May have occasional arrests in movement.
3 = Severely impaired. Frequent hesitation in initiating movements or arrests in ongoing movement.
4 = Can barely perform the task.

Rapid Alternating Movements of Hands

Left upper extremity
Right upper extremity

0 = Normal.
1 = Mild slowing and/or reduction in amplitude.
2 = Moderately impaired. Definite and early fatiguing. May have occasional arrests in movement.
3 = Severely impaired. Frequent hesitation in initiating movements or arrests in ongoing movement.
4 = Can barely perform the task.

Leg Agility

Left lower extremity
Right lower extremity

0 = Normal.
1 = Mild slowing and/or reduction in amplitude.
2 = Moderately impaired. Definite and early fatiguing. May have occasional arrests in movement.
3 = Severely impaired. Frequent hesitation in initiating movements or arrests in ongoing movement.
4 = Can barely perform the task.

Arising from Chair
0 = Normal.
1 = Slow; or may need more than one attempt.
2 = Pushes self up from arms of seat.
3 = Tends to fall back and may have to try more than one time, but can get up without help.
4 = Unable to arise without help.

Posture
0 = Normal erect.
1 = Not quite erect, slightly stooped posture; could be normal for older person.
2 = Moderately stooped posture, definitely abnormal; can be slightly leaning to one side.
3 = Severely stooped posture with kyphosis; can be moderately leaning to one side.
4 = Marked flexion with extreme abnormality of posture.

Gait
0 = Normal.
1 = Walks slowly, may shuffle with short steps, but no festination (hastening steps) or propulsion.
2 = Walks with difficulty, but requires little or no assistance; may have some festination, short steps, or propulsion.
3 = Severe disturbance of gait, requiring assistance.
4 = Cannot walk at all, even with assistance.

Postural Stability – Posture reflexes
0 = Normal.
1 = Retropulsion, but recovers unaided.
2 = Absence of postural response; would fall if not caught by examiner.
3 = Very unstable, tends to lose balance spontaneously.
4 = Unable to stand without assistance.

Body Bradykinesia and Hypokinesia
0 = None.
1 = Minimal slowness, giving movement a deliberate character; could be normal for some persons.
Possibly reduced amplitude.
2 = Mild degree of slowness and poverty of movement which is definitely abnormal.
Alternatively, some reduced amplitude.
3 = Moderate slowness, poverty or small amplitude of movement.
4 = Marked slowness, poverty or small amplitude of movement.

Other questionnaires

Subjective cognitive complaints, Depression, Anxiety, Leisure time activities, Physical activity and Sleep are all assessed with the use of specific questionnaires, shown below.

Subjective Cognitive Complaints Questionnaire

[Vlachos GS, Cosentino S, Kosmidis MH, Anastasiou CA, Yannakoulia M, Dardiotis E, Hadjigeorgiou G, Sakka P, Ntanasi E, Scarmeas N. Prevalence and determinants of subjective cognitive decline in a representative Greek elderly population. Int J Geriatr Psychiatry. 2019 Jun;34(6):846-854. doi: 10.1002/gps.5073. Epub 2019 Mar 22. PMID: 30714214.

Tsapanou A, Vlachos GS, Cosentino S, Gu Y, Manly JJ, Brickman AM, Schupf N, Zimmerman ME, Yannakoulia M, Kosmidis MH, Dardiotis E, Hadjigeorgiou G, Sakka P, Stern Y, Scarmeas N, Mayeux R. Sleep and subjective cognitive decline in cognitively healthy elderly: Results from two cohorts. J Sleep Res. 2019 Oct;28(5):e12759. doi: 10.1111/jsr.12759. Epub 2018 Sep 25. PMID: 30251362; PMCID: PMC6688963.]

1. Is your memory worse than 5 years before?
2. Is your memory worse than that of peers?
3. If yes in 1 or 2, is your memory:
4. If yes in 1 or 2, did the problems with memory start immediately after an illness, an accident or loss of a loved one?
5. Do you find it difficult/hard to remember something you just heard or read?
6. Do you find it hard to remember the meal you had yesterday?
7. Do you need a shopping list to buy what you need, although in the past you did not need one?
8. Does it happen to get lost in your house or not being able to approach a familiar place?
9. Are you able to travel alone (by car, taxi, public transport)?
10. Is it difficult to find the appropriate words while talking, more often than before?
11. Is it difficult to find the names of your relatives and your close friends?
12. Do you take your medication alone without indications?
13. Do you manage the housekeeping like before?
14. Do you have difficulties in calculations when paying (e.g. with change?)
15. Can you sort out your affairs with public services (accountant/tax, bank) without help?

Hospital anxiety and depression scale

[Zigmond AS, Snaith RP. The hospital anxiety and depression scale. Acta Psychiatr Scand. 1983 Jun;67(6):361-70.]

Describe how you felt during the past week

I feel tense or 'wound up'

0=Not at all
1= From time to time, occasionally
2= A lot of the time
3= Most of the time

I still enjoy the things I used to enjoy:

0=definitely as much
1= not quite so much
2= Only a little
3= Hardly at all

I get a sort of frightened feeling as if something awful is about to happen

0= Not at all
1= A little, but it doesn't worry me
2= Yes, but not too badly
3= Very definitely and quite badly

I can laugh and see the funny side of things:

0=as much as always
1= not quite so much
2= definitely not so much now
3= not at all

Worrying thoughts go through my mind

0= Only occasionally
1= From time to time, but not too often
2= A lot of the time
3= A great deal of the time

I feel cheerful:

0=most of the time
1= sometimes
2= not often
3= not at all

I can sit at ease and feel relaxed

0= Definitely
1= Usually
2= Not Often
3= Not at all

I feel as if I am slowed down:

0= Not at all
1= sometimes
2= very often
3= nearly all the time

I get a sort of frightened feeling like 'butterflies' in the stomach:

0= Not at all
1= Occasionally
2= Quite Often
3= Very Often

I have lost interest in my appearance:

0= I take just as much care as ever
1= I may not take quite as much care
2= I don't take as much care as I should
3= Definitely

I feel restless as I have to be on the move

0= Not at all
1= Not very much
2= Quite a lot
3= Very much indeed

I look forward with enjoyment to things:

0= As much as I ever did
1= Rather less than I used to
2= Definitely less than I used to
3= HARDLY AT ALL

I get sudden feelings of panic

0= Not at all
1= Not very often
2= Quite often
3= Very often indeed

I can enjoy a good book or radio or TV program:

0= Often
1= Sometimes
2= not often
3= Very seldom

Leisure Time Activities

[Buchman AS, Boyle PA, Wilson RS, Fleischman DA, Leurgans S, Bennett DA. Association between late-life social activity and motor decline in older adults. Arch Intern Med. 2009;169(12):1139-46.]

Frequency

0=every day or almost every day (5-6-7 times per week)
1=several times a week (3-4 times per week)
2=several times a month (1-5 times per month)
3=several times a year (<once per month)
4=once a year or less
7= NOT INTERP/ DON'T KNOW
8= REFUSED
9= NOT ASKED/ NOT APPLIC

Frequency of participation during last year

1. Whether he knits, plays a musical instrument or deals with another hobby
2. If he goes out walking (for pleasure) or in order to exercise
3. Whether he participates in any physical activity (sports, swimming, gym)?
4. Whether he goes out in order to visit friends or relatives
5. Whether relatives or friends visit the participant at home
6. Whether he goes out to a cinema, a restaurant or in order to participate in any sporting activity
7. Whether the participant reads any magazine
8. Whether he reads newspapers
9. Whether he reads books
10. Whether he watches television
11. Whether he listens to the radio
12. Whether he offers volunteer work
13. Whether he plays cards, games or bingo, whether he bets, or plays chess, backgammon, crosswords
14. Whether he goes to a cafeteria- an open care center for seniors or in another center where he participates in activities
15. Whether he attends any classes
16. Whether he goes to the church

[Buchman AS, Boyle PA, Wilson RS, Fleischman DA, Leurgans S, Bennett DA. Association between late-life social activity and motor decline in older adults. Arch Intern Med. 2009;169(12):1139-46.]
[Mendes de Leon CF, Glass TA, Berkman LF. Social engagement and disability in a community population of older adults: the New Haven EPESE. Am J Epidemiol. 2003 Apr 1;157(7):633-42.]

17. If the participant goes out shopping.
18. If he takes care of the garden.
19. Whether he prepares a meal.
20. Whether he participates in daily trips or in trips that he has to stop over.
21. Whether he participates in different groups of people (e.g. clubs)
22. Whether he continues to have a job for which he is paid

Cognitive activities

[Wilson RS, Bennett DA, Beckett LA, Morris MC, Gilley DW, Bienias JL, Scherr PA, Evans DA.Cognitive activity in older persons from a geographically defined population. J Gerontol B Psychol Sci Soc Sci. 1999;54(3):P155-60.]

23. How often do you visit museums?

Sleep

[Hays, R. D., & Stewart, A. L. (1992). Sleep measures. In A. L. Stewart & J. E. Ware (eds.), Measuring functioning and well-being: The Medical Outcomes Study approach (pp. 235-259), Durham, NC: Duke University Press.]
[Hays RD, Martin SA, Sesti AM, Spritzer KL. Psychometric properties of the Medical Outcomes Study Sleep measure. Sleep Med. 2005;6(1):41-4.]

How long did it take for you to fall asleep during the past 4 weeks?

1=0-15 minutes
2=16-30 minutes
3=31-45 minutes
4=46-60 minutes
5=More than 60 minutes
7= too impaired to respond
8= refused
9= not asked

On average, how long did you sleep each night during the past 4 weeks? (sleep duration)

2a=……..hours
2b=………minutes

(Next questions)

1= All of the time
2= Most of the time
3= A good bit of the time
4= Some of the time
5= A little of the time
6= None of the time
7 =too impaired to respond
8 = refused
9 = not asked

The past 4 weeks,

…... How often did you feel your sleep was not quiet (moving restlessly, feeling tense, speaking) while sleeping?

…... How often did you get enough sleep to feel rested upon waking in the morning?

…... How often did you awake short of breath or with a headache?

…... How often did you feel drowsy or sleepy during the day?

…... How often did you have problem falling asleep?

…... How often did you awaken during your sleep time and have trouble falling asleep again?

…... How often did you have trouble staying awake during the day?

…... How often did you snore during your sleep?

…... How often did you take naps (5 minutes or longer) during the day?

…... How often did you get the amount of sleep you needed?

Questionnaire of Physical Activity (HAPAQ)

Think of the **last 7 days (week)**. We want you to give us some information about your physical activity.

UNIT 1: PHYSICAL ACTIVITY AT WORK			
	What is your main job?		
	Have you worked the last 7 days? If no go to unit 2	0=no	1=yes
	if yes, how many days?		
	How many hours per day on average?		Hours/ day
	From which how much time on average did you consume….		
	Sitting?		Hours/ day
	Standing?		Hours/ day
	In motion?		Hours/ day
	carrying weight?		Hours/ day
	How much time did you need to go and get back from your work?		Minutes/day
	From this time how many minutes…		
	did you walk?		Minutes/day
	did you drive?		Minutes/day
UNIT 2: PHYSICAL ACTIVITY AT HOME			
	During the last 7 days how many hours (on average) **per day**		
	Did you sleep (including siesta)?		hours/day
	Did you watch television-video?		hours/day
	During the last 7 days how many hours <u>**in total**</u> did you consume:		
	For light works at home (eg cooking, washing dishes etc)?		hours/week
	For heavy works at home (eg handwash, mopping etc)?		hours/week
	For reading and computer (beside worktime)?		hours/week
UNIT 3: PHYSICAL ACTIVITY AND ENTERTAINMENT			
	During the last 7 days how many hours <u>**in total**</u>		
	Did you dance in club or/and bar?		hours/week
	Were you sitting or standing with friends in coffee shop-bar-restaurant-theater-cinema?		hours/week
	Did you walk for entertainment (window shopping, park etc) and for transport (beside transport to and from work)?		hours/week
	Have you done any exercise the last 7days	0=no	1=yes
	If yes what exactly did you do and how many hours **in total** during the last 7 days?		
	Type of exercise	Hours/week	
	1.		
	2.		
	3.		
	Main transportation during the last week		
	0= Motorbike, 1=car, 2= walking, 3=bike, 4=public transport, 5=taxi		

Anthropometrics (Height/Weight)

Weight (in kilos) is measured a scale (Model Tefal, ref: PP1133V0 (https://www.tefal-me.com/en/Home-Comfort/TEFAL-CLASSIC-/p/2100098649)

Height (in cm) is measured with a portable stadiometer (Model: seca213, details can be found in: https://www.seca.com/en_ee/products/all-products/product-details/seca213.html).

If objective measurements are not feasible, participants self-report their height and weight.

Dynamometer

Physical strength of the upper extremities is estimated using a handgrip electronic dynamometer (model MG4800, UK). Maximum isometric strength of the dominant hand is recorded three times with an interval of 30 s between each trial, with the participant in the standing position and the elbow at the 90° angle without touching the body.

Gait Speed

Physical performance of the lower extremities is assessed with the 4-m measured walk. After a demonstration, we instruct participants to walk a 4-m distance at their normal walking speed. Each participant has two timed trials to complete the 4-m distance. For each trial, we record the time (in seconds) needed to complete the course.

EEG

EEG recordings are acquired with a 10/20 international system of electrode placement method. All recordings are made with a Micromed SAM 25FO 32-channel headbox (Micromed, Italy) at a sampling rate of 1024 Hz (analogue high pass filter 0.15 Hz) with scalp gold electrodes. A linked earlobe is used as a reference. A 15-min awake resting-state EEG recording is obtained including alternating epochs of eyes closed (EO) and eyes open (EC), each epoch lasting 30 s. Participants are encouraged not to think about anything specific during recording. If participants show signs of drowsiness, they are prompted.

EEG data are analyzed in two ways: (a) spectral analysis of posterior dominant rhythm (PDR) for 3 s after each eye closure. This is a standard analysis, expected to show a gradual decrease in frequency, amplitude, and stability of the PDR with the progression of Alzheimer's disease. (b) a novel non-linear measure of eyes open versus eyes closed EEG synchronization. For this analysis, we are collaborating with the University of Sheffield, which has developed this method.